Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS with 2+ years building Python/TypeScript systems, LLM-backed workflows, enterprise-style integrations, and customer-facing AI features that ship into real production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **LLM-powered grading workflows** handling 10,000+ daily requests, combining OCR, file ingestion, prompts, and REST APIs to turn unstructured submissions into reliable structured outputs for users.
- Worked with users to debug **workflow and execution defects** across upload, persistence, OCR, and grading failures, documenting fixes and reducing feedback turnaround by **60%** during launches.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **customer-facing platform services** for 500+ users, translating stakeholder requests into API changes, data updates, validation logic, and production releases across cloud-hosted workflows and systems for teams.
- Improved **backend workflow performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting production handoffs and debugging work.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and reports for reproducible experiments, anomaly reviews, and model behavior analysis at scale for researchers daily.
- Documented **prompt, evaluation, and reporting workflows** for researchers, turning ad hoc experiments into repeatable processes with clearer technical handoffs, validation outputs, and review notes for teams consistently.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed customer data workflows and operational reporting needs.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions across shared environments and releases consistently for teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **REST and WebSocket services** with SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, integrations, and operational workflows for high-volume users and customers daily reliably together internally.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting, and implementation handoffs during scheduled releases and validations.

Skills

Languages: **Python**, **TypeScript**, JavaScript, SQL, Java, C++, Bash, HTML/CSS
AI Systems: LLM prompts, OCR workflows, model evaluation, RAG-style retrieval, AI-assisted coding
Backend: REST APIs, Flask, FastAPI, Node.js, WebSockets, integrations, async workflows, object-oriented design
Cloud/Infra: AWS Glue, S3, Azure SQL, PostgreSQL, MySQL, MongoDB, Docker, Jenkins, CI/CD
Delivery: Git/GitHub, code reviews, debugging, technical documentation, stakeholder communication, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a **semantic document platform** over 50K+ documents with retrieval, summarization, and contextual Q&A workflows that supported secure, enterprise-style document analysis for teams and analysts internally.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and business data workflows through Java services and TypeScript interfaces for analysts and operators.

HMIS Live | *MERN, ABDM APIs, Sentry, Twilio*

- Implemented **secure operational workflows** integrating patient, provider, claims, notifications, and monitoring features with consent-based APIs, structured validation, and alert-driven reliability across teams and systems during daily operations.